multiaccess media

shared air for radio/light signals

shared wires for electrical/light signals

...

needs:

- way to tell what signals are for whom
- way to decide who 'talks' when/where

shared wires

used to be how Ethernet worked

- before Ethernet switches were ubiquitous

how cable Internet works

- shared line to many customers in area
- need to share

usually how fiber-to-the-home works

- “passive optical network”
- connect multiple fibers optically

multiple channels

can have multiple channels over single medium

typically: radio or electrical signal or light frequencies

sender/receiver separate channels electrically/optically

to start: will worry about coordinating one chnanel

wireless spectrum

most useful radio spectrum *licensed*

gov't gives exclusive rights (within some region) to specific organizations/poeple

- example: cellular, TV, satellite, air traffic control, etc.

a lot of computer networking uses *unlicensed* bands

- use without specific permission allowed
- still limits on power, procedures to avoid interference

selected unlicensed bands

approx. frequencies unlicensed in US (not everywhere)

902–928 MHz (802.15.4 (IoT focused))

2.4–2.5 GHz (802.15.4; 802.11b/g/n/ax/...; bluetooth; microwave ovens)

5.15 GHz–5.25 GHz (802.11a/n/ac/ax/...)

5.25 GHz–5.73 GHz (802.11a/n/ac/ax/...; also weather radar)
requires 'dynamic frequency selection'

5.73 GHz–5.85 GHz (802.11a/n/ac/ax/...)

5.93 GHz–7.12 GHz (802.11ax/...)

collisions

N nodes try to transmit one channel at same time

likely outcomes for some receiver:

receiver gets garbage

"collision"

receiver receives 1 of the N collisions

running example

based on Abramson, "The Aloha System—Another alternative for computer comunications" (1970)

suppose we have shared radio with nodes A1, A2, ..., An and B

A1, A2, ...An are all trying to transmit to B

takes 1 ms to send message

and want to collectively send k messages per second

randomly spaced (exponential distribution)

some probability

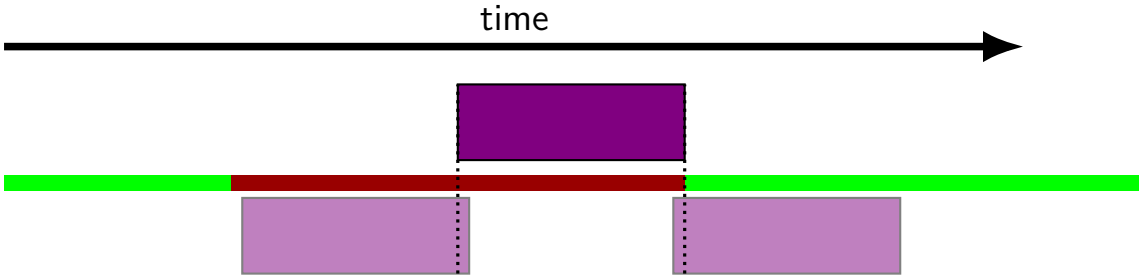
exponential distribution with mean λ

- our model for when packets sent
- "memoryless" distribution
- knowing when last packet sent tells you nothing about next
- (yes, not realistic)

probability events occur $< K$ time units apart

$1 - e^{-\lambda K}$

quiet time to avoid collisions



to avoid collision with 1 ms packet...

can't start packet less than 1 ms before

can't start packet less than 1 ms after

$\rightarrow$ need 2 ms without packet starting for no collision

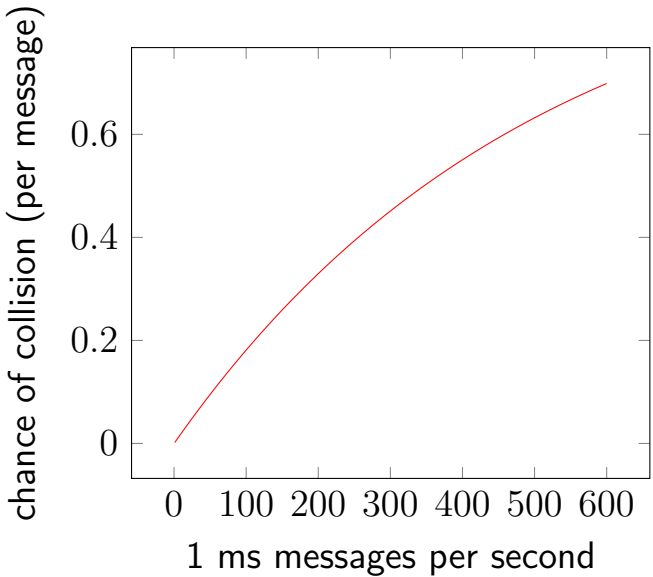
chance of collisions? (1)

to avoid collision when sending 1 ms packet

need no other packet to be sent in 2ms period around its start time

with k packets/sec, chance is approx $1 - e^{-\frac{2}{1000}k}$

chance of collision



retransmissions

what's going to happen when node can't send message

probably it will retransmit it...

which means real transmission rate will be some $R > k$

where k is rate messages are generated

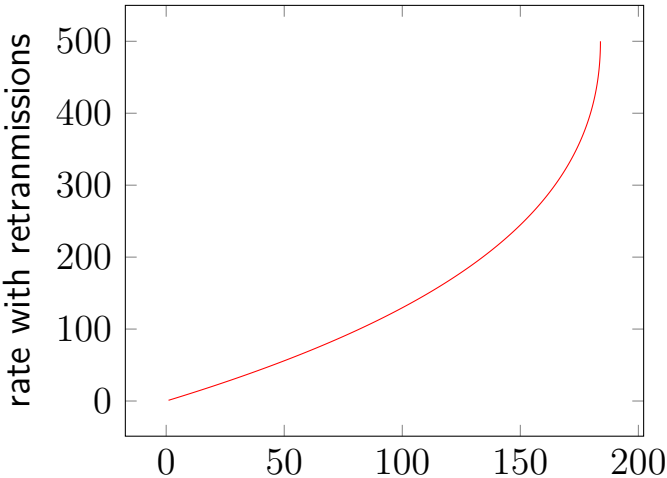
about $[1 - e^{-2R\frac{1}{1000}}]$ chance of each message generated

so $R = k + \left(1 - e^{-2R\frac{1}{1000}}\right) \cdot R$

$R = k + R - Re^{-2R\frac{1}{1000}}$

$k = Re^{-2R\frac{1}{1000}}$

retranmissions (plot)



thinking about result

sending 500 1ms packet or retransmission/second

- using about half the capacity!

representing $\sim$ 186 1 ms non-retranmissions/second

- $\frac{1}{2e} = 0.186\ldots$
- using about 1/6th the capacity

results hold generally

seems pretty bad for shared channel efficiency!

carrier sense

channel can be ‘busy’ or not

radio/light:

> have some sort of signal detectable on frequency

“carrier sense”

way to detect whether channel busy

using carrier sense

simple idea: don't transmit if channel already busy

problem: then when do you transmit?

some options:

- never, lose the packet
- immediately when it stops being busy
- sometime after it stops being busy

‘slots’

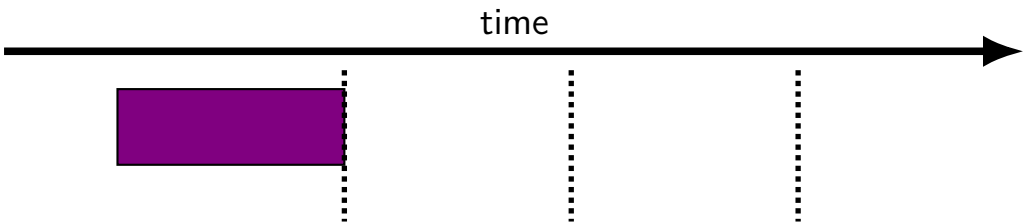
if packets take a fixed amount of time...

want to wait whole number of packet times

immediately after, one after, two after, three after

avoids requiring almost 2 packets worth of ‘quiet’ time

can synchronize by observing end of last transmission



802.11b typical slots

20μs slots

50μs interframe spacing (IFS)

includes time to receive ACK of packet

transmit first time after IFS (if idle)

transmit second time after IFS + rand$(0, 2^5)$ slots

plus time spent by other seen transmissions

third time after rand$(0, 2^6)$ slots, etc.

collision detection

Wi-Fi: use ACKs to detect if transmission successful

problem: transmit whole packet, then know about collision

alternate idea: listen for collision, stop transmitting early

doesn't work on wireless (we'll talk about why later)

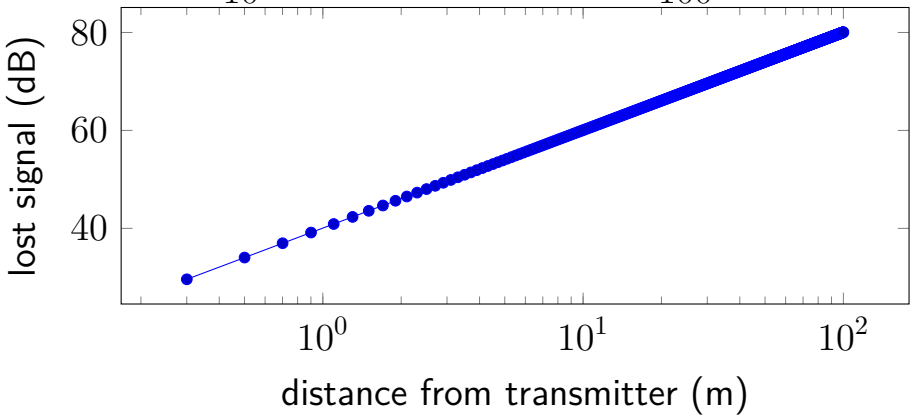
but does work on some wired shared media

part of reason for design of Ethernet header

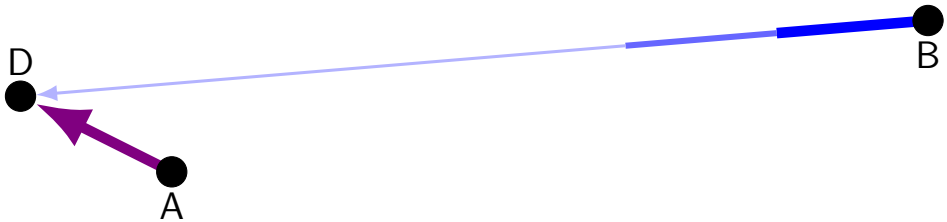
free space wireless transmission

assuming 2.4GHz Wifi, no obstacles, omnidirectional transmission:

10 dB loss $= \frac{1}{10}$th power; 20 dB loss $= \frac{1}{100}$th power



receiving multiple nodes

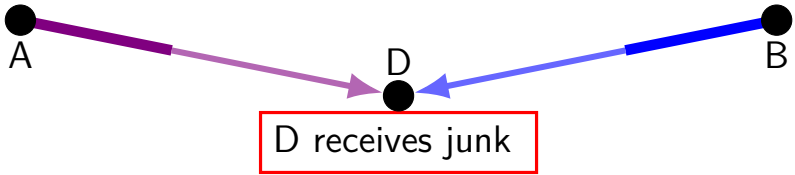


A's signal much much stronger at D

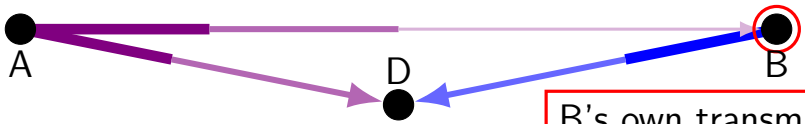
D may still receive A's signal...

because B relatively too weak to interfere

collision (non-)detection

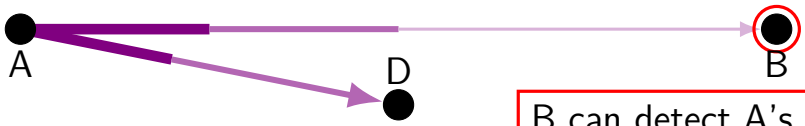


collision (non-)detection



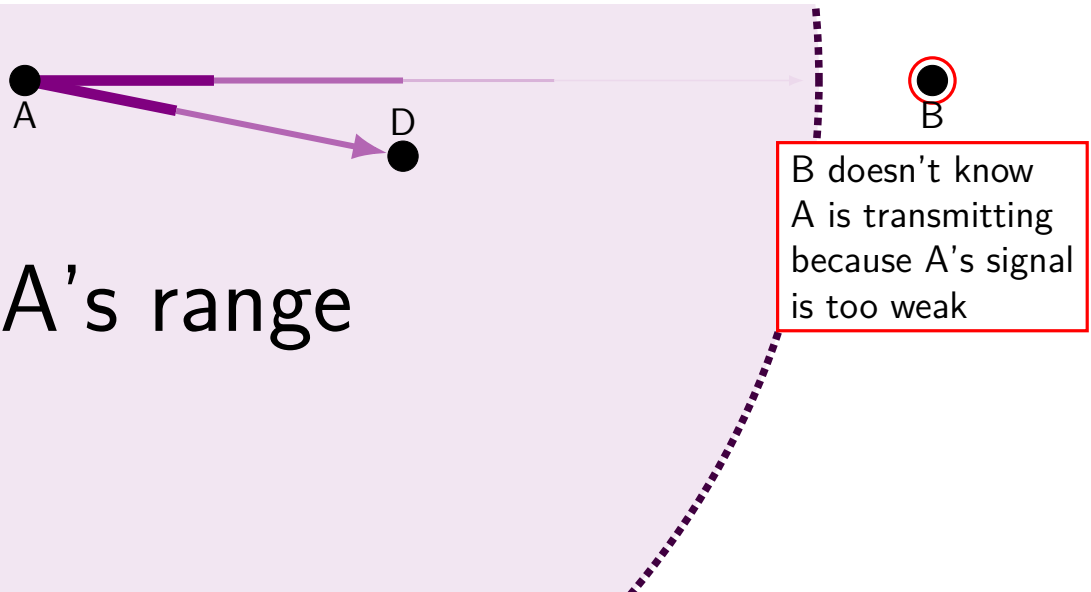
B's own transmission
is much stronger than A's
cannot detect collision itself

carrier-sense

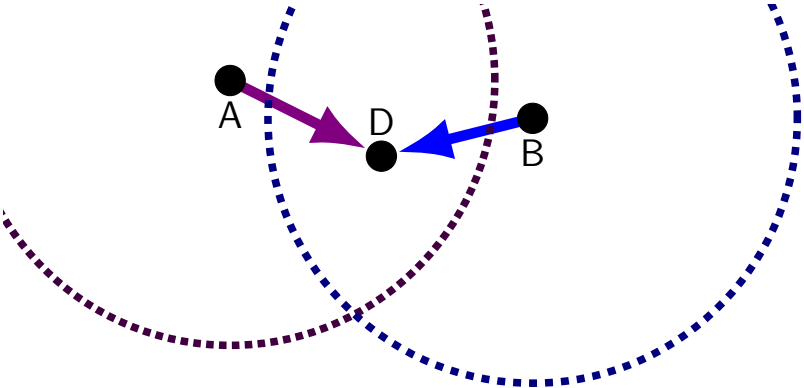


B can detect A's transmitting
avoid starting transmission now

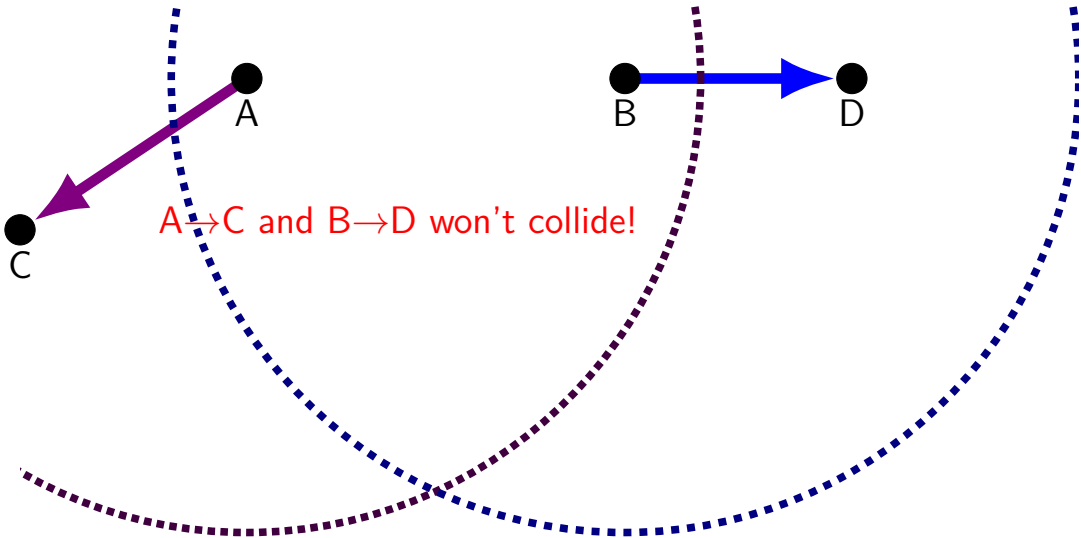
carrier-not-sensing



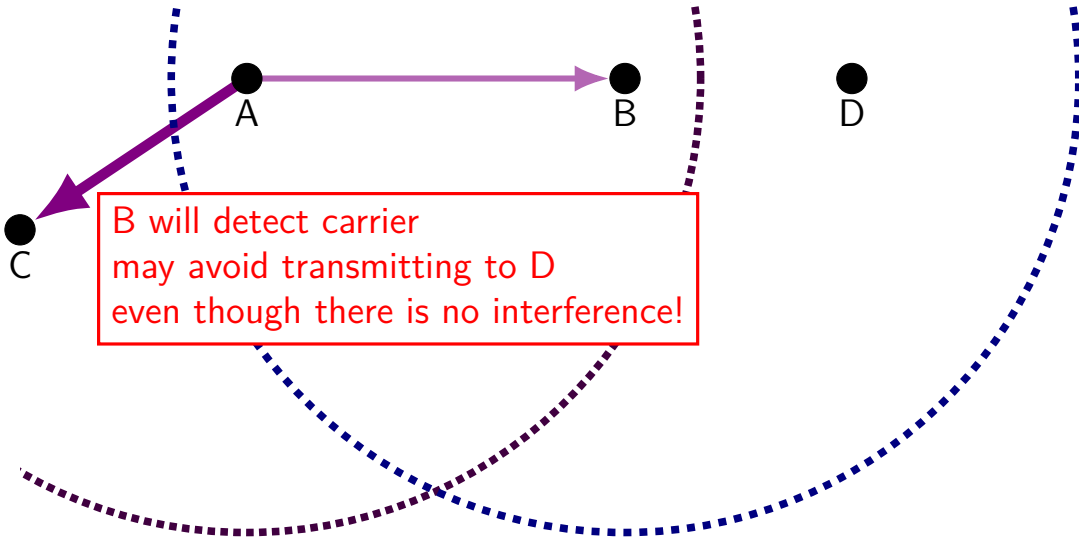
‘hidden node’ problem



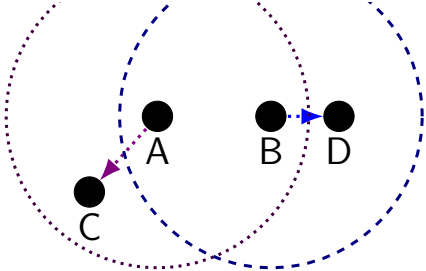
carrier false sensing



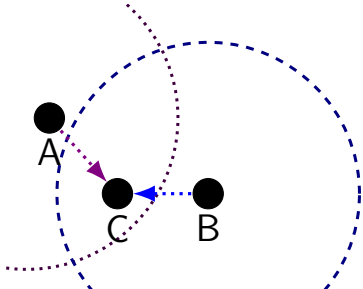
carrier false sensing



hidden/exposed node summary



exposed node problem: A and B detect each other's transmissions but don't interfere



hidden node problem: A and B interfere when sending to C but can't detect each other's transmissions

request-to-send/clear-to-send

can't detect likely interference at sender

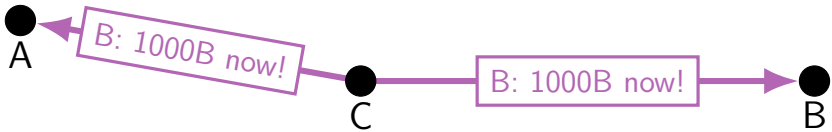
idea: have receiver tell when it's okay to transmit

sender→receiver: request-to-send

receiver→sender: clear-to-send



overhearing clear-to-send

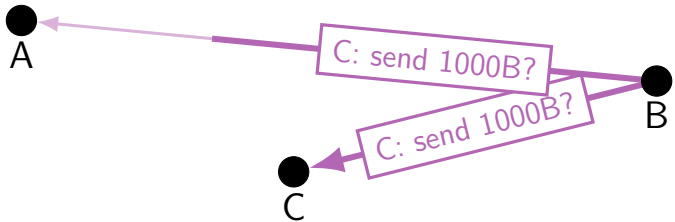


A can infer that B is transmitting to C

...and for how long!

A can avoid interfering, even if out of range of carrier-sense

overhearing ready-to-send



if A can hear B's request-to-send...

A should avoid sending until after B should get clear-to-send

exercise: request-to-send advantage

assume:

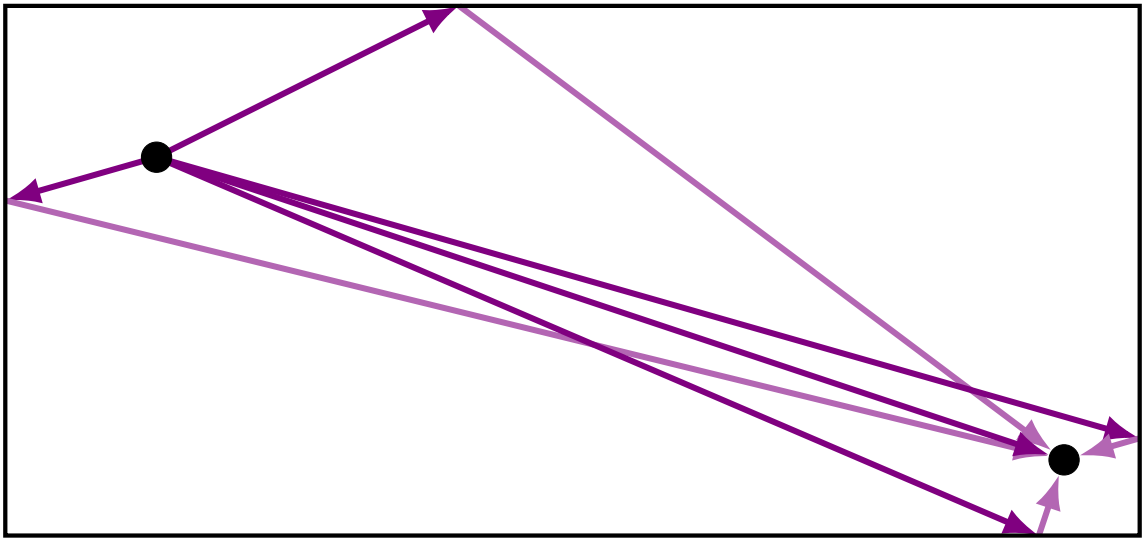
- 1μs request-to-send/clear-to-send messages
- 100μs messages sent
- chance of collision proportional to message lengths

exercise: effect of using RTS/CTS (versus only carrier-sense) for...

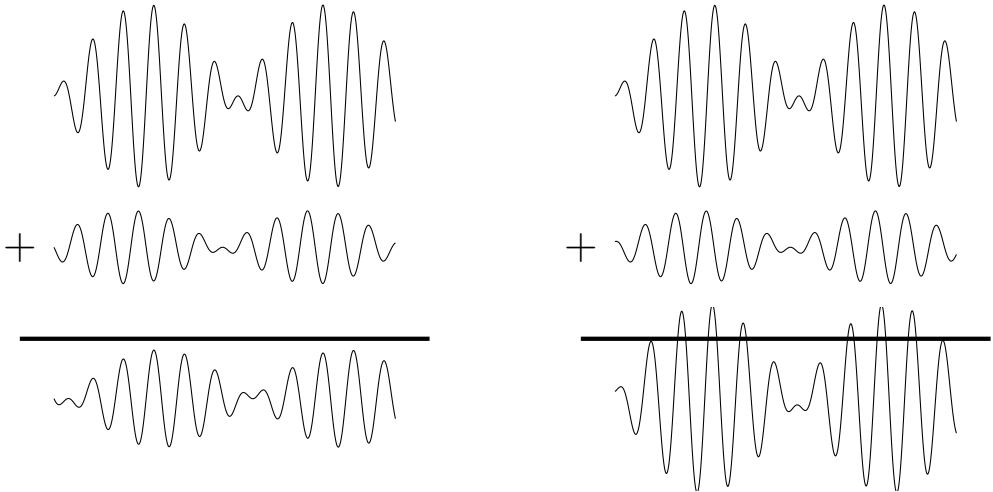
2 nodes transmitting to one with no hidden nodes?

2 nodes hidden from each other transmitting to one?

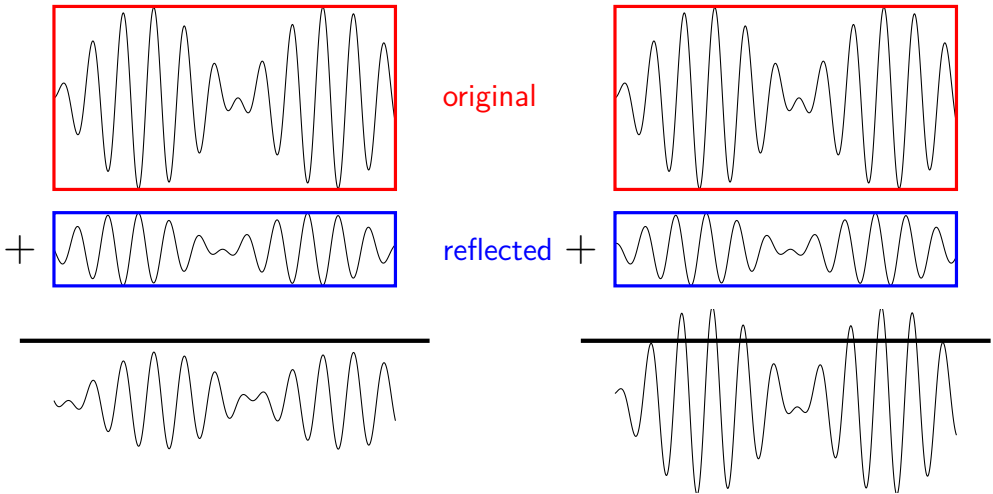
multipath interference



simple self-interference



simple self-interference



multipath interference

quite complicated to predict in three dimensions, variable reflectivity, ...

can help and hurt signal reception

contributes to irregular dead zones, etc.

deliberate multipath

can have:

multiple sending antennas (inputs)

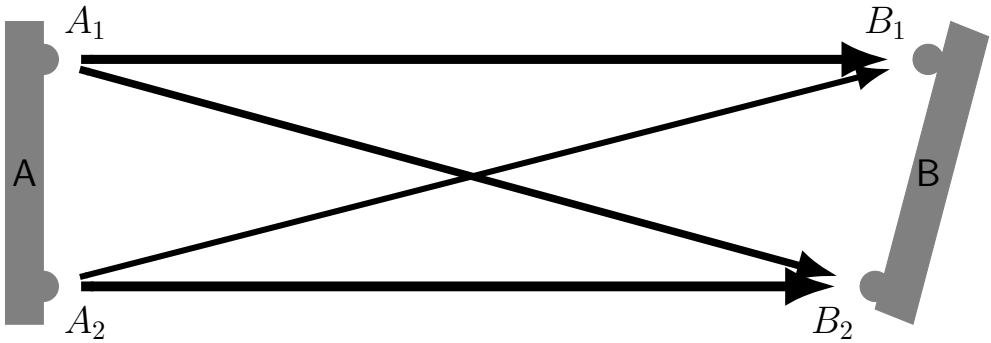
multiple receiving antennas (outputs)

some simple ideas:

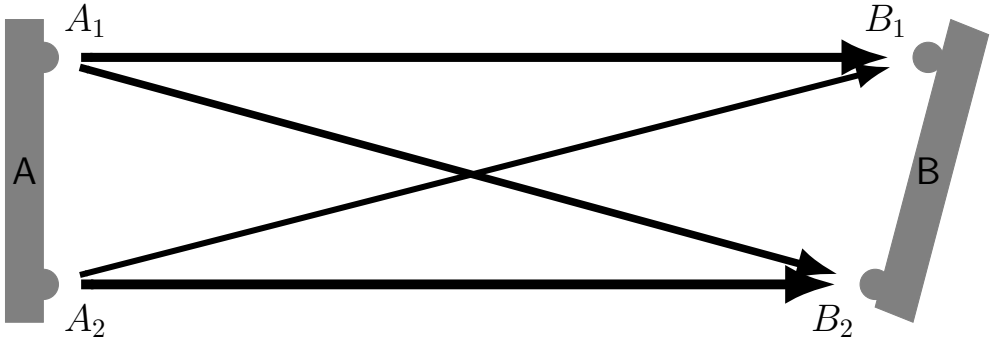
receive signal twice, hope one antenna not dead zone

send signal twice, have receiver tell you which is better

multiple spatial streams



multiple spatial streams



signals might mix based on relative strengths, say

$$B_1 \sim 0.9A_1 + 0.7A_2$$
$$B_2 \sim A_1 + 1.2A_2$$

simplified multipath model

$$
\begin{aligned}
B_1 \sim \quad & 0.9A_1 + \quad & 0.7A_2 \\
B_2 \sim \quad & A_1 + \quad & 1.2A_2
\end{aligned}
$$

if we can estimate coefficients...

B can solve for A_1, A_2

A can send with pattern to help estimate coefficients:

example: send full power from A_1 ($(B_1, B_2) = (0.9, 1)$), then from A_2 ($(B_1, B_2) = (0.7, 1.2)$)

correlation problem

when two 'spatial streams' correlated, harder to solve for them

example: suppose $(B_1, B_2) = (A_1 + A_2, A_1 + 1.1A_2)$

$A_2 = 10B_1 - 10B_2$

need 10x precision in B_1, B_2 to get 1x precision in A_2

wifi modes

"ad-hoc": everyone sends/receives from whoever

"infrastructure": designated *access points*

mostly use infrastructure networks

infrastructure mode

networks identified by SSID (Service Set IDentifier)

- human readable name

single SSID can have many access points (example: eduroam)

clients *associated* with one access point at a time

clients only *only* send to their current access point

...even if sending to other nodes on network

beacons (approx. once per 100ms)

```
▸ Frame 1074: 433 bytes on wire (3464 bits), 433 bytes captured (3464 bits) on interface mon0, id 0
▸ Radiotap Header v0, Length 54
▸ 802.11 radio information
▾ IEEE 802.11 Beacon frame, Flags: ........
    Type/Subtype: Beacon frame (0x0008)
  ▸ Frame Control Field: 0x8000
    .000 0000 0000 0000 = Duration: 0 microseconds
    Receiver address: ff:ff:ff:ff:ff:ff
    Destination address: ff:ff:ff:ff:ff:ff
    Transmitter address: d8:07:b6:d9:[redacted]
    Source address: d8:07:b6:d9:[redacted]
    BSS Id: d8:07:b6:d9:[redacted]
    .... .... .... 0000 = Fragment number: 0
    1010 0000 0000 .... = Sequence number: 2560
    [WLAN Flags: .......]
▾ IEEE 802.11 Wireless Management
  ▸ Fixed parameters (12 bytes)
  ▾ Tagged parameters (343 bytes)
    ▸ Tag: SSID parameter set: "[redacted]"
    ▸ Tag: Supported Rates 6(B), 9, 12(B), 18, 24(B), 36, 48, 54, [Mbit/sec]
    ▸ Tag: DS Parameter set: Current Channel: 157
    ▸ Tag: Traffic Indication Map (TIM): DTIM 0 of 1 bitmap
    ▸ Tag: Country Information: Country Code US, Environment All
    ▸ Tag: RSN Information
    ▸ Tag: Vendor Specific: Microsoft Corp.: WPA Information Element
    ▸ Tag: RM Enabled Capabilities (5 octets)
    ▸ Tag: Supported Operating Classes
    ▸ Tag: HT Capabilities (802.11n D1.10)
    ▸ Tag: HT Information (802.11n D1.10)
    ▸ Tag: Extended Capabilities (6 octets)
    ▸ Tag: Interworking
    ▸ Tag: Advertisement Protocol
    ▸ Tag: VHT Capabilities
    ▸ Tag: VHT Operation
    ▸ Tag: Tx Power Envelope
    ▸ Ext Tag: HE Capabilities
    ▸ Ext Tag: HE Operation
    ▸ Ext Tag: MU EDCA Parameter Set
    ▸ Ext Tag: Spatial Reuse Parameter Set
    ▸ Tag: Vendor Specific: Microsoft Corp.: WMM/WME: Parameter Element
    ▸ Tag: Vendor Specific: Microsoft Corp.: WPS
    ▸ Tag: Vendor Specific: Wi-Fi Alliance: Multi Band Operation - Optimized Connectivity Experience
    ▸ Tag: Vendor Specific: Metalink LTD.
```

probes/probe responses

beacons — listen for a few seconds, find out about nearby networks

probably need to scan multiple channels

can also send explicit probes to learn about networks

receive responses, with essentially same kind of information

association

client → AP: association response (SSID=...)

AP → client: association response (SSID=...)

+ things related to WiFi Security (encryption, passwords, etc.)

client → AP: deassociation (sometimes)

wifi data frames

```
▸ Frame 2389: 233 bytes on wire (1864 bits), 233 bytes captured (1864 bits) or
▸ Radiotap Header v0, Length 68
▸ 802.11 radio information
▾ IEEE 802.11 QoS Data, Flags: .p....F.
    Type/Subtype: QoS Data (0x0028)
  ▸ Frame Control Field: 0x8842
    .111 1110 1100 1111 = Duration: 32463 microseconds
    Receiver address: a0:b3:39:43:e8:fb
    Transmitter address: d8:07:b6:d9:ae:4e
    Destination address: a0:b3:39:43:e8:fb
    Source address: d8:07:b6:d9:ae:50
    BSS Id: d8:07:b6:d9:ae:4e
    STA address: a0:b3:39:43:e8:fb
    .... .... .... 0000 = Fragment number: 0
    0001 1110 1111 .... = Sequence number: 495
    [WLAN Flags: .p....F.]
  ▸ Qos Control: 0x0000
  ▸ CCMP parameters
▸ Data (131 bytes)
```

wifi data

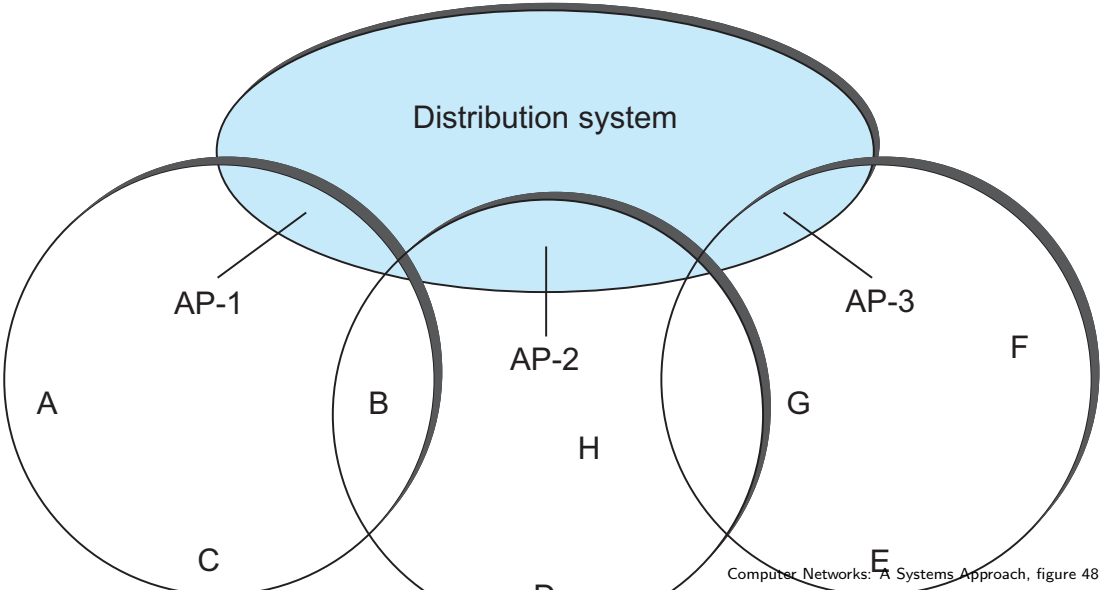
receiver/transmitter address — this hop

source/destination address — final soruce/destination

can differ if...

- destination is a broadcast/multicast address
- wireless equivalent of 'switching'

multiple APs, one network



APs as switches

multiple APs can act as *one network*

example: nodes connected to different APs can send packets to each other by MAC address

distribution system needs to share neighbor-table-like information

switching APs

nodes can switch APs...

check for new APs when signal strength too low

periodically check for beacons + signal strength

switching APs = same as joining network, but...

- can keep IP address, etc.
- maybe optimizations to avoid redoing wifi security, etc.

Roofnet (Cambridge, MA, 2005)

Bicket, Aguayo, Biswas, Morris, "Architecture and Evaluation of an Unplanned 802.11b Mesh Network (MobiCom'05)



Figure 1: A map of Roofnet. Each dot represents a wireless node. The map covers the south-east portion of Cambridge, Massachusetts. The Charles River curves along the lower boundary of the map, MIT is at the lower right, and Harvard is at the upper left.

roofnet routing

up to five hops for packet to get from one node to another

link-state routing protocol between nodes

- no explicit configuration needed

sending nodes computed list of hops + included in packet

- idea called "source routing"
- prevents transient routing loops
- important because conditions (e.g. weather) changes connectivity

everyone using same channel!

link-state/distance vector for wireless

no explicit list of links/networks

instead: periodically broadcast and see who responds

keep track of signal strength/reliabliy/etc. for routing metrics

modern mesh networks

common for distribution network between APs to be (partly) wireless

sysadmin view:

- plug some APs into internet connection
- put other APs in appropriate place
- APs figure out how to make it work

typically using self-organizing mesh network ideas

- likely similar routing protocols to what we discussed

usually properietary networking protocols

- vendor lock-in problem
- (though there is now a recent Wifi standard)

deliberate scheduling

so far:

- devices decide when to try to send
- channel supporting one transmitter

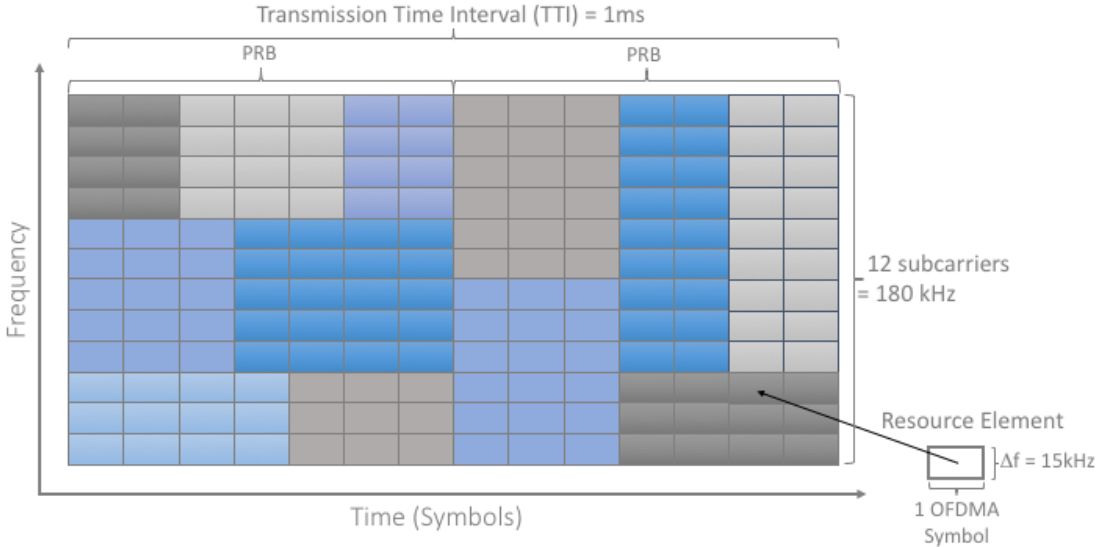
alternate idea (cellular networks): divide

several available channels

- typically different frequencies
- signal encoding chosen to avoid interference between close frequencies

slots within each time interval

Computer Networks: A Systems Approach, Figure 54



central scheduler idea

'base stations' track active user devices

users have bandwidth reservations

base stations send out schedule of slots every $\sim$ 0.1–1ms

protocol for handoff between base stations

devices send back quality feedback to aid scheduling

exercise

central scheduling v carrier-sense

exercise: which handles better...

- utilizing the most of the available bandwidth
- independently controlled access points/base stations
- communications between two nearby nodes

backup slides